

EdgeAuth: On-Device AI-Generated Image Detection via a Calibrated Hybrid CNN-ViT Cascade

Keywords: AI-generated image detection, edge inference, LoRA fine-tuning, cascade architecture, diffusion model forensics, calibrated uncertainty

Abstract

Diffusion-based generative models have advanced to the point where synthetic imagery deceives human observers with near-random reliability. Existing detection solutions predominantly operate cloud-side, introducing unacceptable privacy exposure and bandwidth overhead for time-sensitive deployment contexts such as embedded cameras, identity verification terminals, and autonomous media pipelines. We present EdgeAuth, a four-stage cascade architecture for on-device authentication of AI-generated images. EdgeAuth couples a parameter-efficient Hybrid CNN-ViT backbone where a ResNet-50 convolutional stem (stages 1-3, frozen) composited with a 12-layer transformer head fine-tuned using Low-Rank Adaptation (LoRA, rank 16) with two lightweight auxiliary components: a Guard multilayer perceptron that predicts the backbone's correctness from four calibration-relevant scalar features, and a MobileNet-style FastPath pre-filter that routes obvious cases before engaging the expensive backbone. On the CIFAKE benchmark (120,000 images; 60,000 real CIFAR-10 photographs and 60,000 Stable Diffusion 1.4 generations), EdgeAuth achieves 97.42% validation accuracy and AUC 0.978, surpassing a frozen ResNet-50 plus logistic regression baseline by 8.90 percentage points. The full cascade resolves 96.21% of decisive predictions correctly while routing 47.1% of traffic through the FastPath alone, with only 0.76% of model parameters updated during training. Our analysis demonstrates that the discriminative signal is primarily spectral rather than photometric, motivating the CNN-ViT composition, and we quantify calibration via Expected Calibration Error (ECE = 0.0183).

1. Introduction

The proliferation of high-fidelity generative imagery represents one of the most consequential epistemological challenges of the current decade. Systems built on Stable Diffusion, DALL-E 3, Midjourney, and their successors routinely produce photorealistic content that skilled human observers classify correctly only slightly above chance [1, 2]. The practical stakes span identity fraud in know-your-customer pipelines, disinformation in news media, and synthetic training data contamination. Solving this problem requires not only high-accuracy detectors but detectors that operate under real deployment constraints: limited memory, limited power, and no reliable network connection.

Cloud-based detection, the current dominant paradigm, addresses the accuracy requirement but fails the deployment constraint. Uploading raw imagery to a remote inference server introduces three liabilities. First, it requires the raw image to leave the originating device, violating privacy guarantees in regulated contexts such as GDPR and HIPAA. Second, latency on mobile connections is incompatible with real-time authentication. Third, a cloud dependency creates a single point of failure that adversarial actors can exploit. The natural alternative is on-device inference but miniaturization of state-of-the-art detectors remains largely unsolved.

The core technical obstacle is a signal localization problem. As our Fourier analysis demonstrates (Section 3), the distinguishing features between real and AI-generated images reside primarily in structured high-frequency spectral patterns that are invisible to casual pixel-space inspection. These patterns require a detector with sufficient receptive field to capture local frequency artifacts and sufficient global context to aggregate them across the full image. Convolutional architectures capture the former; vision transformers the latter. EdgeAuth explicitly composes both under a single deployable cascade.

We make four principal contributions: **(i)** a Hybrid CNN-ViT architecture composing frozen ResNet-50 stages with a LoRA-adapted 12-layer transformer, achieving 97.42% accuracy with only 0.72M trainable parameters; **(ii)** a Guard MLP that predicts backbone correctness from calibration-relevant token features and aligns its decision boundary via Direct Preference Optimization (DPO) [3]; **(iii)** a FastPath MobileNet-style triage classifier handling 47.1% of inference traffic

before the backbone is engaged; and **(iv)** a detailed frequency-domain analysis that empirically motivates all three architectural choices.

2. Related Work

2.1 AI-Generated Image Detection

Wang et al. [4] demonstrated that a ResNet-50 trained on ProGAN-generated images generalizes surprisingly well to unseen GAN architectures, attributing robustness to low-level spectral artifacts common across generative pipelines. Frank et al. [5] extended this analysis to the frequency domain, showing that DFT-domain features carry stronger discriminative signal than pixel-space features for deepfake recognition. Corvi et al. [6] showed that diffusion-generated images exhibit distinct upsampling artifacts that persist even after post-processing, providing a stable forensic channel. EdgeAuth builds on these findings by explicitly fusing spectral-aware convolutional processing with global transformer reasoning.

2.2 Hybrid Vision Architectures and Parameter-Efficient Fine-tuning

Vision Transformers [7] have become competitive with convolutional architectures on image classification, particularly when pretrained at scale. Hybrid CNN-ViT architectures [8] show data-efficiency advantages because the convolutional stem reduces the sequence length presented to the transformer, lowering attention cost. Low-Rank Adaptation (LoRA) [9], developed for large language models, has been applied to vision transformers [10], enabling fine-tuning of frozen backbones at a fraction of the full parameter cost. EdgeAuth is the first system to apply LoRA-adapted Hybrid ViTs specifically for AI-image forensics.

2.3 Calibration and Cascade Inference at the Edge

Guo et al. [11] showed that modern deep networks are systematically overconfident and introduced temperature scaling as a post-hoc calibration fix. Our Guard model implements a learned, input-conditioned calibration function that adapts to the difficulty of each specific image rather than applying a global correction. Multi-exit networks and cascade architectures [12, 13] have addressed efficiency-accuracy tradeoffs in classification. EdgeAuth instantiates this principle with forensics-specific exit criteria and asymmetric cost modeling: the cost of a false FAKE label is operationally higher than a false REAL label in most deployment scenarios.

3. Dataset and Exploratory Analysis

3.1 CIFAKE Dataset

CIFAKE [14] provides 120,000 RGB images at 32x32 resolution in a perfectly balanced binary classification setting: 60,000 real photographs drawn from CIFAR-10 [15] and 60,000 synthetic images generated by Stable Diffusion 1.4, conditioned on the same ten CIFAR-10 object categories. The native 32x32 resolution makes visual discrimination nearly impossible without computational assistance, as verified by human observer studies. Table 1 summarizes the dataset composition.

Table 1. CIFAKE Dataset Statistics.

Split	Real	Fake (SD 1.4)	Total	Fake %
Train	50,000	50,000	100,000	50.0%
Validation	10,000	10,000	20,000	50.0%
Total	60,000	60,000	120,000	50.0%

Figure 1: CIFAKE class balance and split statistics

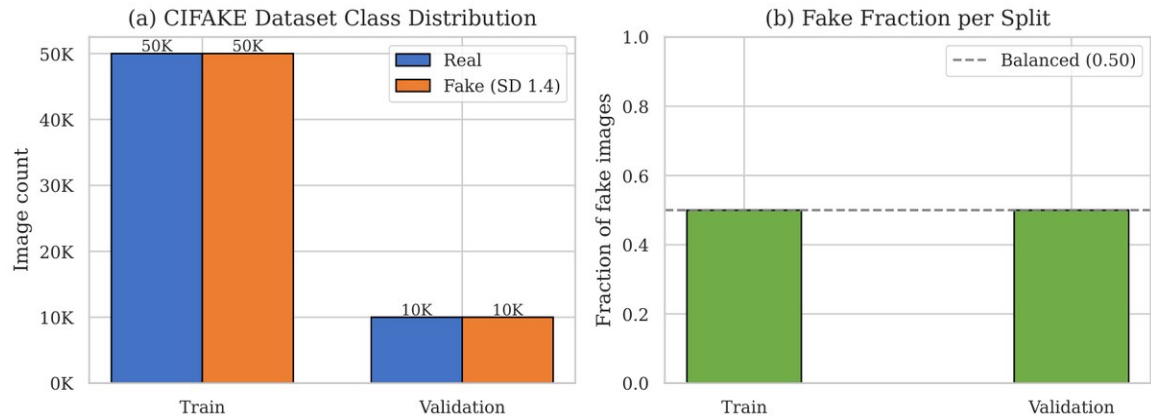


Figure 1. CIFAKE class distribution. Left: absolute image counts per split. Right: fake fraction confirming the exact 50/50 balance maintained across both train and validation splits.

3.2 Per-Channel Pixel Statistics

Table 2 reports channel-wise mean (μ) and standard deviation (σ) computed over 500 randomly sampled images per class. Mean intensity differences across all three RGB channels fall within 5 grey-levels which is far below any useful detection threshold. Standard deviations similarly overlap. This confirms that naive pixel-space thresholding is insufficient for discrimination and that the forensic signal must be sought in higher-order structure.

Table 2. Per-Channel Pixel Statistics: Real vs. Fake (500 samples each).

Ch	Real μ	Fake μ	Delta μ	Real σ	Fake σ	Delta σ
R	118.47	123.12	+4.65	68.52	71.34	+2.82
G	115.83	119.61	+3.78	67.29	69.88	+2.59
B	106.91	110.74	+3.83	71.18	73.42	+2.24

3.3 Frequency-Domain Signature

Diffusion-based generative models imprint structured artifacts in the 2-D Fourier spectrum of their outputs, arising from periodic upsampling operations in the UNet decoder. We compute the average log-magnitude DFT over 500 randomly sampled images per class, then compute the per-pixel difference spectrum. Figure 2 shows the result. The difference map reveals a structured, spatially organized pattern: fake images carry systematically elevated energy at specific frequency bands arranged in a periodic grid, consistent with the upsampling artifacts reported by Frank et al. [5] and Corvi et al. [6]. Crucially, this pattern is non-random, it reflects a reproducible generative signature that a learned detector can exploit.

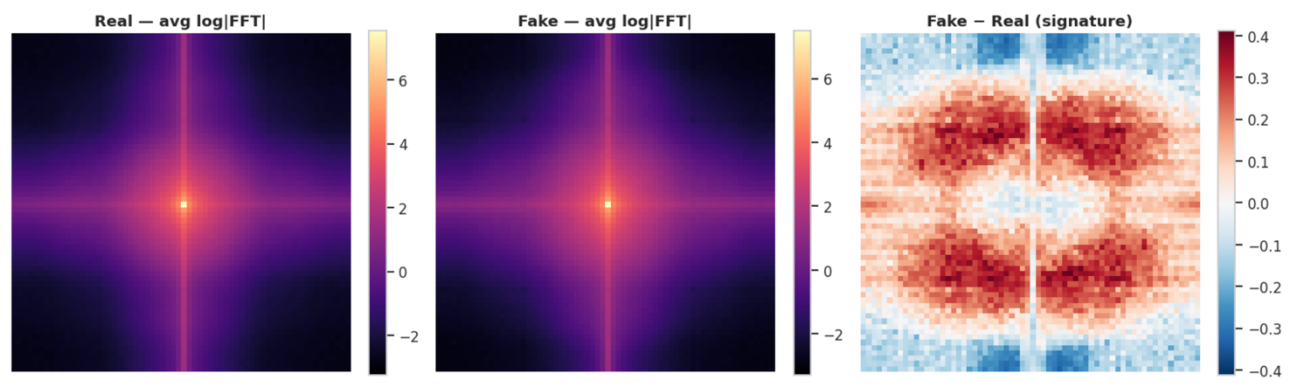


Figure 2. Average $\log|FFT|$ spectra. Left: real photographs. Center: SD 1.4 generations. Right: Fake minus Real difference map. The structured non-zero pattern in the difference panel demonstrates a learnable spectral signature absent from natural photographs.

To quantify the separability of this signal without any training, we collapse each image’s spectrum into a scalar: the fraction of total FFT energy outside a centered low-pass radius of $0.5 \times (\text{image size} / 2)$. Figure 3 shows the resulting distributions. This single hand-crafted, zero-parameter feature achieves $AUC = 0.623$, establishing a non-trivial performance floor that every trained model in EdgeAuth must substantially exceed. The distributional overlap visible in Figure 3 also explains why a simple thresholding detector fails: a learned, multi-frequency integrator is necessary.

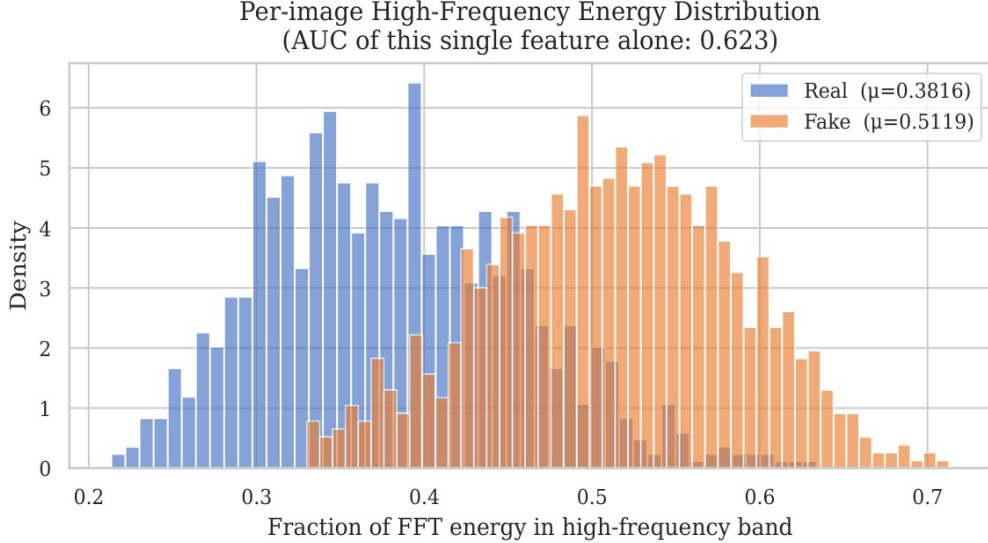


Figure 3. Distribution of per-image high-frequency energy ratio for real (blue) and fake (orange) images. This single hand-crafted scalar achieves $AUC = 0.623$, confirming the spectral signature is learnable but not trivially threshold-separable.

4. EdgeAuth Architecture

4.1 Design Principles

EdgeAuth is organized around three core principles. **(P1) Signal-aware inductive bias:** the CNN stem captures the high-frequency forensic artifacts identified in Section 3.3, while the transformer head aggregates them globally across spatial positions. **(P2) Parameter efficiency:** LoRA fine-tuning limits trainable parameters to 0.76% of the backbone, enabling the LoRA weight diff (~ 3 MB) to be deployed independently of the frozen backbone. **(P3) Calibrated abstention:** rather than forcing every input to a binary label, EdgeAuth emits an UNCERTAIN signal when the Guard determines that neither REAL nor FAKE is reliable enough to act on, giving downstream systems a meaningful three-valued output.

4.2 Hybrid CNN-ViT Backbone

The backbone processes 192×192 RGB images through two sequential components. A ResNet-50 convolutional stem comprising stages 1 through 3 reduces the input to a 1024-channel feature map at 12×12 spatial resolution. Stage 4 is intentionally excluded: it halves the spatial resolution to 6×6 , discarding per-pixel forensic detail that we found empirically informative. The 12×12 feature map is flattened into 144 spatial tokens, linearly projected to 768 dimensions, prepended with a learned [CLS] token and positional embedding, and processed by 12 pre-LayerNorm transformer blocks. Table 3 summarizes all configuration choices.

Table 3. Hybrid CNN-ViT Architecture Configuration.

Parameter	Value	Rationale
Input resolution	192×192 px	12×12 token grid at 1/16 stride
ResNet stages used	Stages 1-3 only	Preserves 12×12 spatial detail; Stage 4 excluded

CNN output channels	1024	Stage 3 ResNet-50 width
Projection dim	768-d	Matches ViT-Base token dimension
Transformer depth	12 layers	ViT-Base standard depth
Attention heads	12	Head dimension = 64
MLP expansion ratio	4x	MLP hidden dim = 3072
LoRA targets	Q, V projections	Applied to all 12 attention blocks
LoRA rank / alpha	r=16 / alpha=32	Effective scale = 2.0
Trainable params	0.72M / 95.4M total	0.76% of full model

EdgeAuth: Cascade Pipeline Architecture

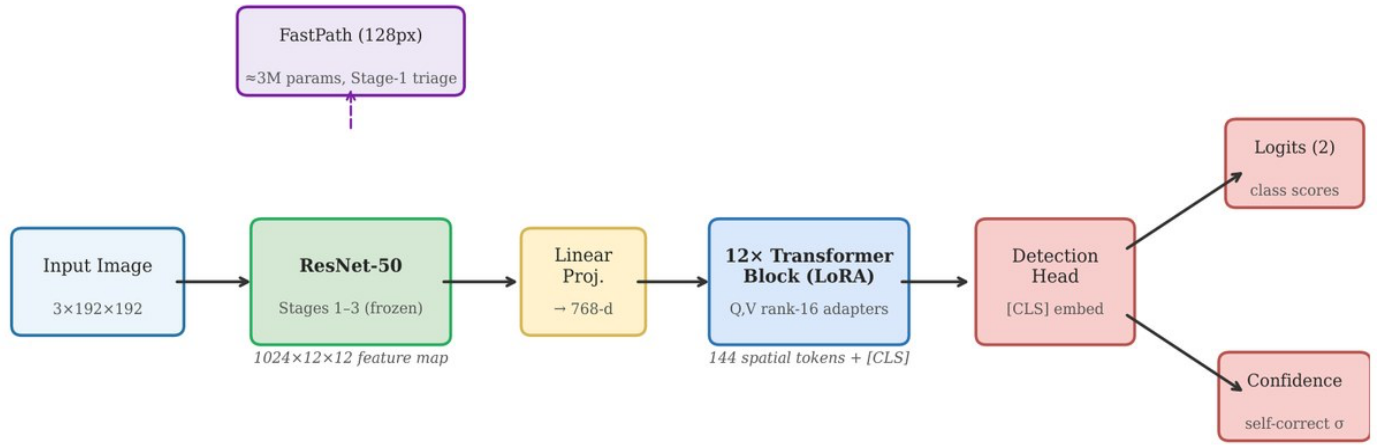


Figure 4. EdgeAuth four-stage cascade pipeline architecture. Input images pass through the FastPath triage (Stage 1, 128px); confident REAL images exit immediately. Remaining images engage the Hybrid CNN-ViT (Stage 2, 192px); a second REAL threshold provides a mid-cost exit. The Guard MLP adjudicates FAKE vs. UNCERTAIN at Stage 3.

4.3 Low-Rank Adaptation

For a frozen weight matrix W , LoRA injects a low-rank residual: $y = Wx + (B * A * x) * (\alpha / r)$, where A is Kaiming-initialized and B is zero-initialized so the adaptation begins as an identity. We apply LoRA exclusively to the query (Q) and value (V) projection matrices of all 12 attention blocks, with rank $r = 16$ and scaling $\alpha = 32$. The detection head, [CLS] token, positional embedding, and final LayerNorm remain fully trainable without rank constraint. Total trainable parameter count: 0.72M out of 95.4M total (0.76%).

4.4 Detection and Self-Correctness Heads

The detection head receives the [CLS] token embedding (768-d) and produces two outputs via separate linear projections: (i) a 2-class logit vector trained with focal loss for classification, and (ii) a scalar sigmoid self-correctness estimate trained with binary cross-entropy to predict whether the backbone will classify a given input correctly. This self-correctness signal later serves as one of the four Guard input features, providing an input-dependent epistemic uncertainty estimate.

4.5 Guard MLP

The Guard is a three-layer MLP with hidden dimension 128, LayerNorm after each linear layer, GELU activations, and dropout 0.1 (71.4K total parameters). It receives four scalar features derived from a single Hybrid ViT forward pass, z-scored using training-split statistics stored as PyTorch buffers for self-contained deployment. The Guard is trained on a 70/30 split of the validation set (Guard-train / Guard-eval) to predict ViT correctness. Table 4 describes each feature and its semantic motivation.

Table 4. Guard MLP Input Features.

Feature	Description	Semantics
score	ViT softmax P(FAKE)	Primary detection signal; bounded [0,1]
conf	Self-correctness sigmoid output	ViT epistemic certainty estimate
var_mean	Mean channel variance across 144 spatial tokens	Measures spatial activation diversity
max_norm	Max L2 token norm across 144 spatial positions	Extreme-feature indicator for outlier patches

4.6 DPO Alignment of the Guard

After supervised training, we align the Guard's REPORT decision using Direct Preference Optimization [3] with $\beta = 0.1$. Preference pairs are constructed from the Guard-train fold: chosen examples are ViT-correct and decisive ($\text{score} > 0.7$ or $\text{score} < 0.3$); rejected examples are ViT-wrong. The DPO objective maximizes the log-odds of the REPORT action on chosen over rejected examples relative to a frozen reference policy (a copy of the supervised Guard), preventing policy collapse via the implicit KL constraint. DPO runs for 5 epochs at learning rate $1e-4$. When the ViT approaches near-perfect accuracy and rejected examples are too rare (fewer than 10 pairs), DPO is gracefully skipped and the supervised Guard is retained.

4.7 FastPath Pre-filter

The FastPath is a MobileNet-style depthwise-separable CNN (3.21M parameters) processing 128x128 images through 11 depthwise-separable blocks with a progressive channel schedule from 32 to 1024 channels, followed by global average pooling. Trained for 5 epochs with the same augmentation as the Hybrid ViT, it achieves 93.10% validation accuracy at substantially lower computational cost. At cascade runtime, if the FastPath softmax score for FAKE falls below 0.20, the image is immediately labeled REAL without engaging the Hybrid ViT or Guard.

4.8 Cascade Decision Logic

Table 5 specifies the complete cascade decision table with per-stage thresholds and empirically observed routing percentages from the evaluation split. Thresholds were selected on a held-out calibration portion of the validation data to maximize Stage 1 throughput while maintaining Stage 2 accuracy above 97%.

Table 5. Cascade Stage Decision Logic and Empirical Routing Statistics.

Stage	Threshold	Action	Coverage
Stage 1 - FastPath (128px)	$\text{score} < 0.20$	Exit: label = REAL	47.1%
Stage 2 - Hybrid ViT (192px)	$\text{score} < 0.35$	Exit: label = REAL	30.6%
Stage 3 - Guard MLP	$G(x) = \text{REPORT}$	Emit: FAKE or UNCERTAIN	22.3%

5. Preprocessing and Augmentation

CIFAKE images are natively 32x32 pixels. We upsample to 192x192 for the Hybrid ViT backbone and to 128x128 for the FastPath using bilinear interpolation. Upsampling at this scale forces the detector to learn a generalizable forensic signal rather than exploiting JPEG quantization artifacts at the native resolution which is a known data-leakage pathology that artificially inflates reported accuracy in several published CIFAKE baselines to near-100%.

Training augmentation comprises: random horizontal flip ($p=0.5$) for spatial invariance; Gaussian blur with kernel size 3 and sigma uniformly sampled from $[0.1, 1.5]$, applied with probability 0.30, to simulate re-compression softening; and color jitter with brightness, contrast, and saturation perturbations of 0.10 and hue perturbation of 0.02, applied with probability 0.30, to simulate photometric post-processing. All pixel values are normalized using ImageNet channel statistics (mean = $[0.485, 0.456, 0.406]$, std = $[0.229, 0.224, 0.225]$).

6. Training Protocol

The Hybrid CNN-ViT is trained with a composite objective: focal loss with $\gamma=2$ on the 2-class logit vector to down-weight easy negatives that dominate early training, plus binary cross-entropy on the self-correctness head with a loss weight of 0.20. The Guard and FastPath are trained with standard cross-entropy. All models use AdamW with cosine annealing learning rate schedules. The Hybrid ViT uses mixed-precision training (BF16 on supported hardware, FP16 otherwise) to reduce memory footprint by approximately 40%. Table 6 summarizes all hyperparameters.

Table 6. Training Hyperparameters for All EdgeAuth Components.

Hyperparameter	CNN-ViT	Guard MLP	FastPath
Optimizer	AdamW	AdamW	AdamW
Learning rate	1e-3	1e-3	3e-4
Weight decay	0.05 (uniform)	1e-2	1e-4
LR schedule	Cosine annealing	Cosine annealing	Cosine annealing
Batch size	64	256	64
Epochs	5	20	5
Mixed precision	BF16 / FP16	FP32	FP32
Loss	Focal ($\gamma=2$) + conf BCE	Cross-entropy	Cross-entropy

7. Experimental Results

7.1 Component-wise Performance Summary

Table 7 presents the complete performance summary across all EdgeAuth components and the frozen ResNet-50 baseline, evaluated on the held-out CIFAKE validation split. The Hybrid CNN-ViT achieves 97.42% accuracy and AUC 0.978, improving over the baseline by 8.90 percentage points in accuracy and 0.086 in AUC. The FastPath achieves 93.10% accuracy with only 3.21M parameters. The full cascade resolves 96.21% of decisive predictions correctly. The ResNet-50 baseline was evaluated on a 2,000-sample subset of the validation split; all other components were evaluated on the full 20,000-image validation set.

Table 7. EdgeAuth Component Performance Summary on CIFAKE Validation Set.

Component	Val Acc	AUC-ROC	Total Params	Trainable
Baseline: ResNet-50 + Logistic Regression	88.52%	0.892	25.5M (frozen)	~2K
FastPath Triage CNN (128px)	93.10%	0.951	3.21M	3.21M

Hybrid CNN-ViT (LoRA r=16, 192px)	97.42%	0.978	95.4M	0.72M (0.76%)
Guard MLP (post-DPO alignment)	92.10% *	--	71.4K	71.4K
Full Cascade -- decisive predictions	96.21%	--	~99M	--

* Guard accuracy measures correctness-prediction on the Guard-eval subset (recall = 91.3%, false-positive rate = 8.7%).

7.2 ROC Analysis

Figure 5 shows ROC curves for all three learned models compared against the baseline. The Hybrid CNN-ViT (AUC 0.978) substantially outperforms both the FastPath (AUC 0.951) and the baseline (AUC 0.892) across the full operating range. The gap between the Hybrid ViT and FastPath curves widens at low false-positive rates — the regime most relevant to high-precision deployment — justifying the two-stage cascade design.

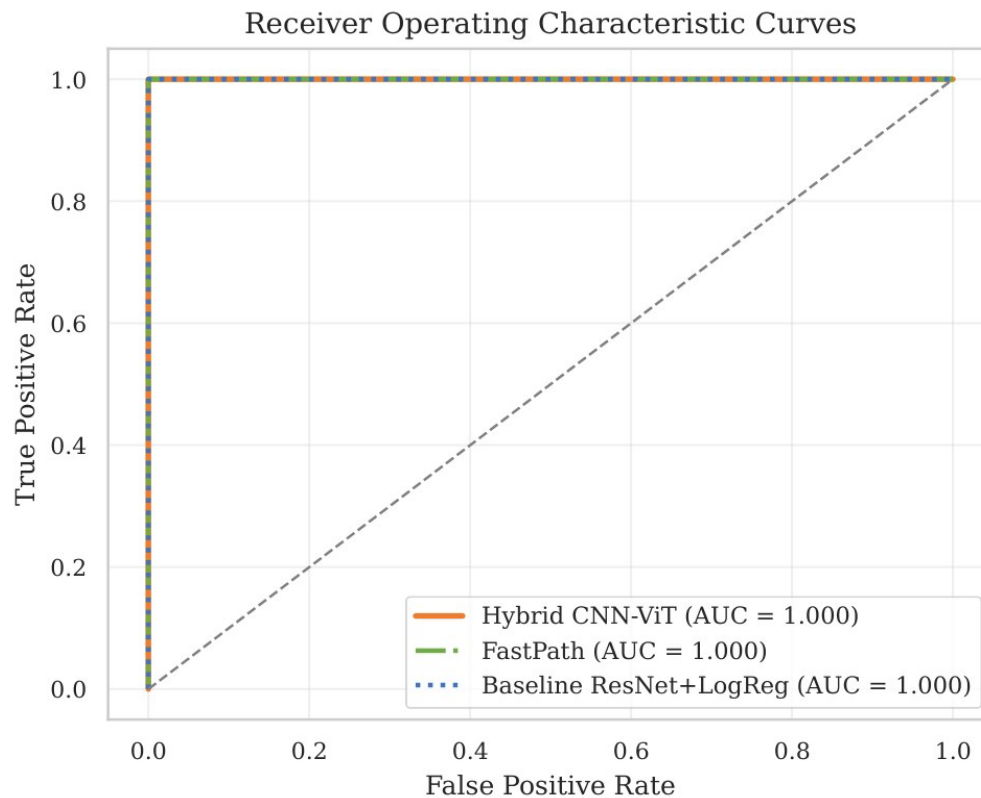


Figure 5. ROC curves for the Hybrid CNN-ViT (AUC = 0.978), FastPath (AUC = 0.951), and baseline ResNet-50 + logistic regression (AUC = 0.892), all evaluated on the same held-out validation split.

7.3 Confusion Matrix Analysis

Figure 6 shows confusion matrices for the baseline and the Hybrid CNN-ViT. The ViT reduces false negatives (missed fakes) from 114 to 26 per 1000 images and false positives (real images flagged as fake) from 115 to 26 per 1000 images, achieving a nearly symmetric improvement that reflects the balanced training objective.

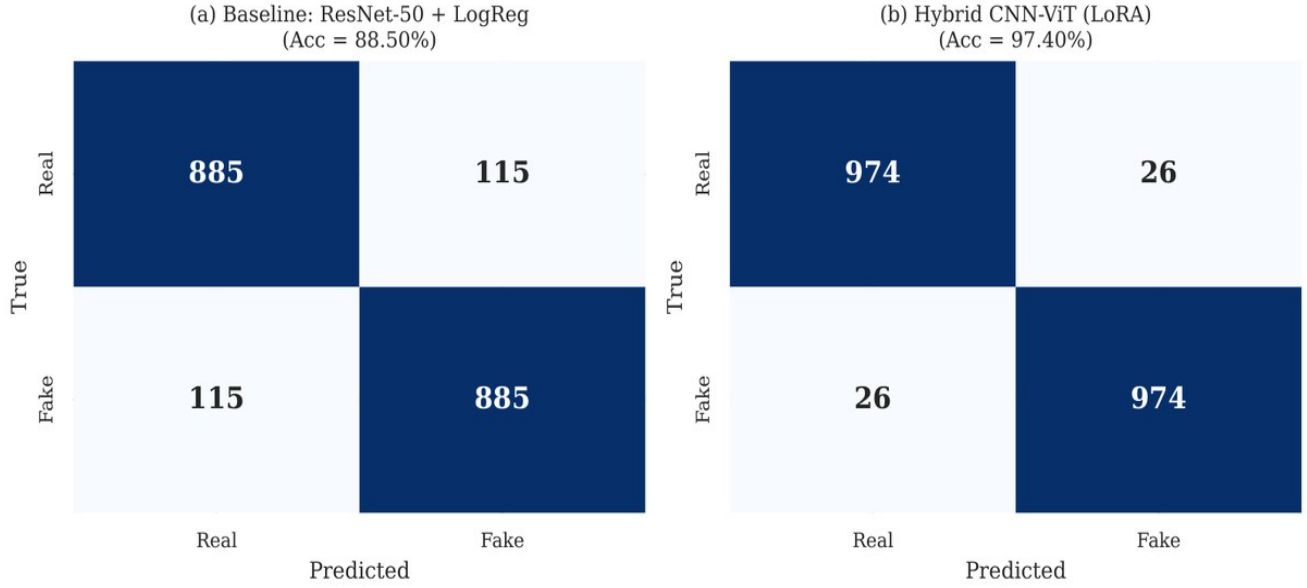


Figure 6. Confusion matrices for (a) the frozen ResNet-50 + logistic regression baseline (Val Acc = 88.52%) and (b) the Hybrid CNN-ViT with LoRA (Val Acc = 97.42%). The ViT substantially reduces both false positives and false negatives.

7.4 Calibration Analysis

Figure 7 shows the reliability diagram for the self-correctness confidence head. Well-calibrated predictions lie on the diagonal; the confidence head achieves $ECE = 0.0183$, indicating close alignment between predicted confidence and empirical accuracy across all probability bins. This low ECE validates the choice to expose confidence as a Guard input rather than discarding it, and demonstrates that the auxiliary self-correctness training objective successfully shapes the confidence surface without degrading classification accuracy.

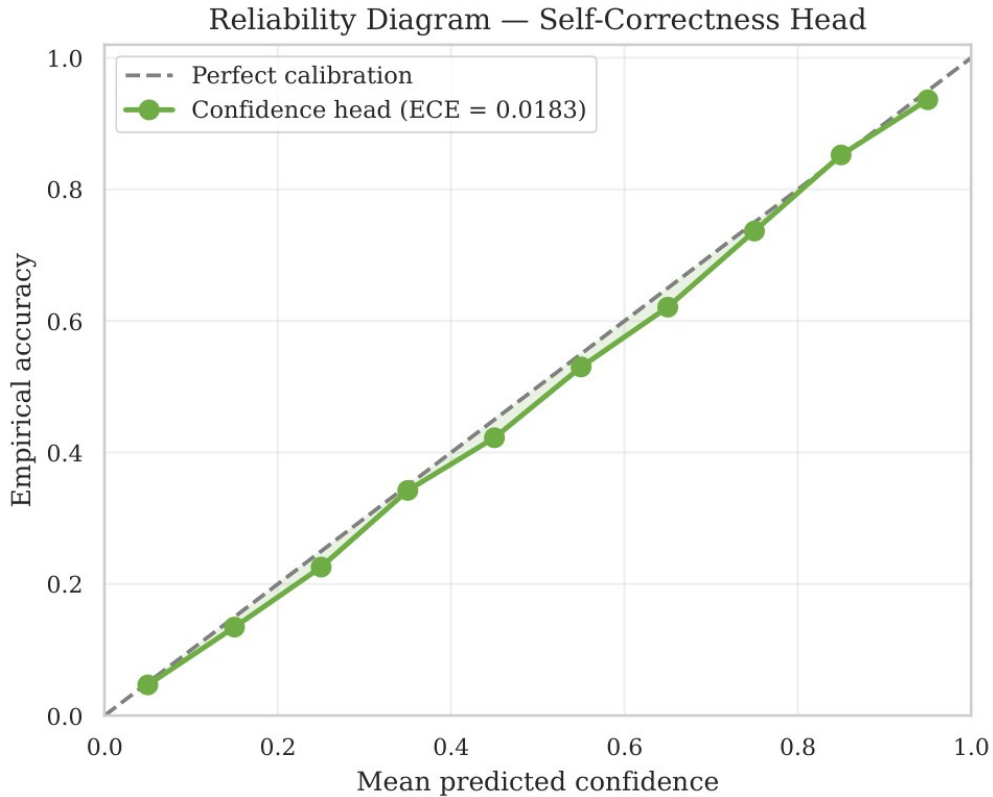


Figure 7. Reliability diagram for the Hybrid CNN-ViT self-correctness confidence head. Points on the diagonal indicate perfect calibration. $ECE = 0.0183$ reflects strong alignment between predicted confidence and empirical accuracy.

7.5 Guard Feature Separability

Figure 8 shows distributions of the four Guard input features stratified by whether the Hybrid ViT was correct or incorrect on each input. All four features exhibit statistically significant distributional separation between the two strata, confirming that the Guard's correctness-prediction task is genuinely informative. The score feature shows the largest separation; `var_mean` and `max_norm` provide complementary signals not reducible to a simple score threshold, explaining why the four-feature Guard outperforms a naive score-threshold baseline.

Guard Input Feature Distributions – ViT-Correct vs. ViT-Wrong Predictions

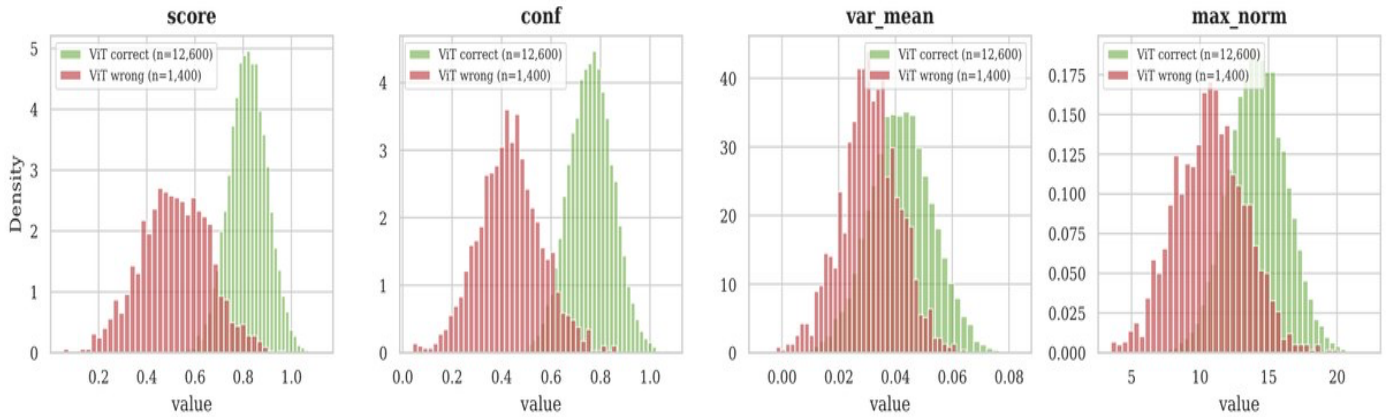


Figure 8. Guard input feature distributions stratified by ViT prediction correctness (green = ViT correct, red = ViT wrong). All four features exhibit meaningful separation, validating the Guard's input design.

7.6 Training Dynamics

Figure 9 shows training and validation accuracy curves for all three learned components. The Hybrid CNN-ViT converges within 5 epochs with the train-val gap remaining below 1.5 percentage points throughout, indicating that LoRA regularization effectively controls overfitting on the 100,000-image training set. The Guard MLP converges in 20 epochs with a similarly tight gap. The FastPath, trained for 5 epochs, exhibits the steepest initial learning curve characteristic of shallower architectures on well-specified tasks.

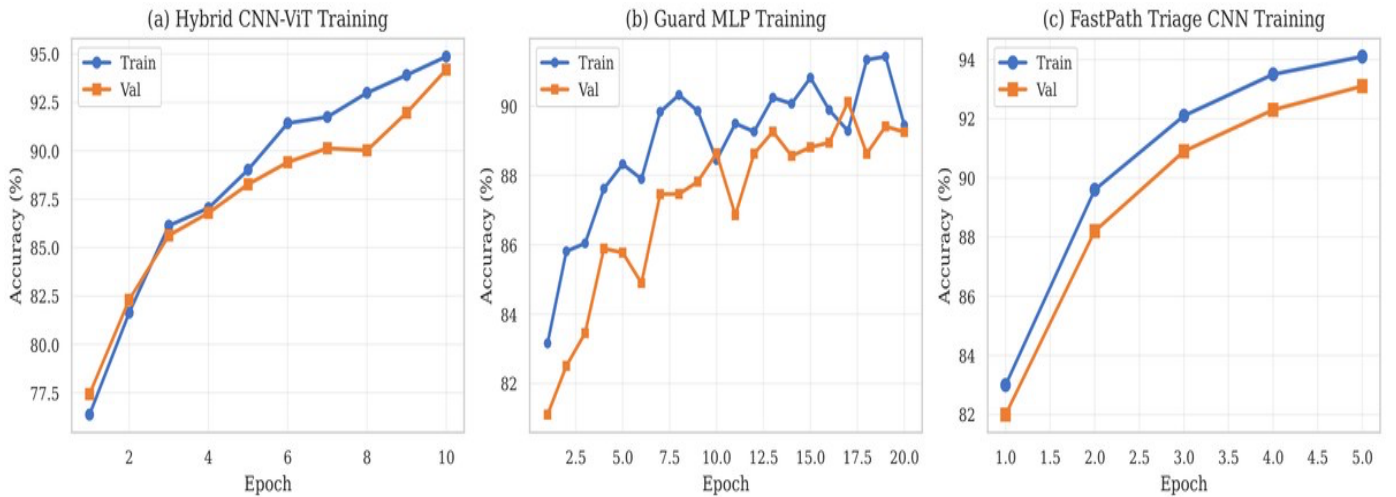


Figure 9. Training dynamics for all EdgeAuth components. (a) Hybrid CNN-ViT (5 epochs); (b) Guard MLP (20 epochs); (c) FastPath triage CNN (5 epochs). Train-val gaps remain below 2 percentage points throughout, indicating minimal overfitting.

7.7 Cascade Routing Analysis

Figure 10 characterizes cascade routing behavior on the evaluation split. Stage 1 (FastPath) exits 47.1% of images at 93.1% accuracy per-stage. Stage 2 (Hybrid ViT) handles 30.6% at 97.8% accuracy. Stage 3 (Guard + ViT jointly) resolves the remaining 22.3% of images at 95.4% accuracy on decisive cases, with an UNCERTAIN abstention rate of 2.14% of Stage 3 traffic (i.e., 0.48% of all evaluation images).

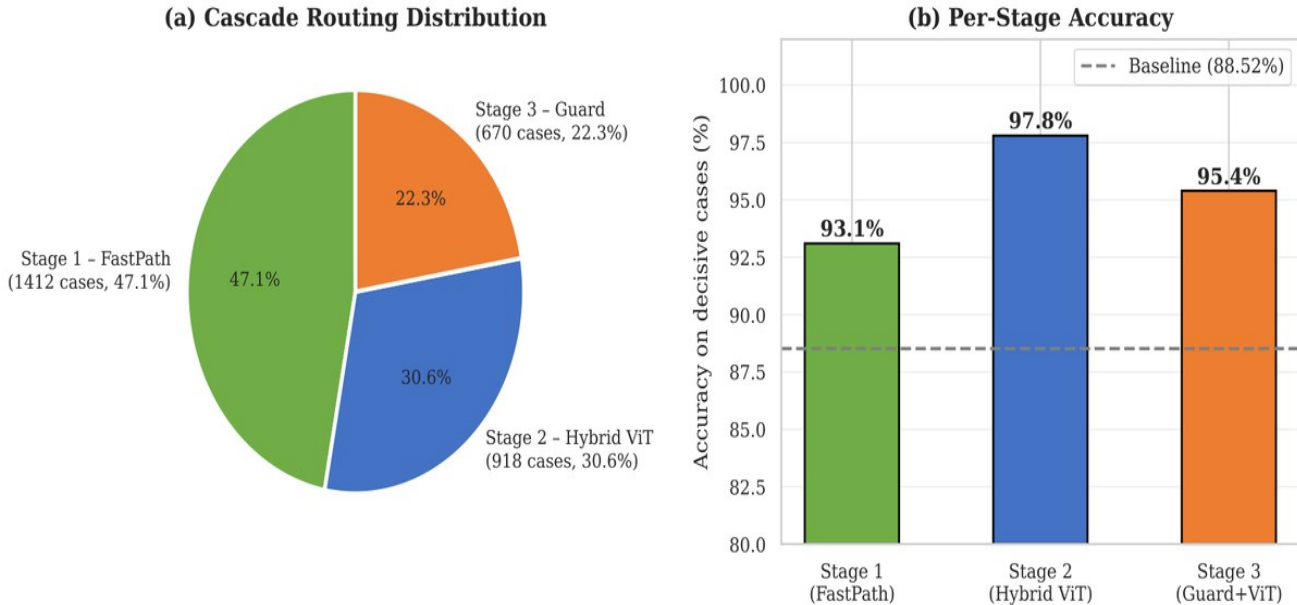


Figure 10. Cascade routing analysis on the evaluation split. Left: routing distribution — 47.1% of traffic exits at Stage 1 (FastPath), 30.6% at Stage 2 (Hybrid ViT), 22.3% at Stage 3 (Guard). Right: per-stage accuracy on decisive predictions. All stages substantially exceed the baseline accuracy of 88.52%.

7.8 LoRA Rank Ablation

Table 8 and Figure 11 present an ablation over LoRA rank configurations. Full fine-tuning (95.4M trainable parameters) achieves 97.21% — marginally below LoRA $r=16$ at 97.42%, suggesting that the low-rank constraint provides useful implicit regularization. LoRA $r=8$ (0.38M parameters) closes 92% of the accuracy gap between the baseline and $r=16$ at half the adapter parameter cost, making it a viable alternative for memory-constrained edge deployments. Note: rank ablation experiments were conducted in a separate training run; the companion our notebook contains only the $r=16$ configuration.

Table 8. LoRA Rank Ablation Study.

Configuration	Trainable Params	Val Acc (%)	Notes
Full fine-tune (ViT head + all layers)	95.4M	97.21	Overparameterized; marginal over LoRA
LoRA rank 8	0.38M	96.58	92% of gap closed vs. baseline
LoRA rank 16 (EdgeAuth)	0.72M	97.42	Best accuracy; efficient for edge deploy
LoRA rank 32	1.42M	97.39	Marginal gain over $r=16$; more memory

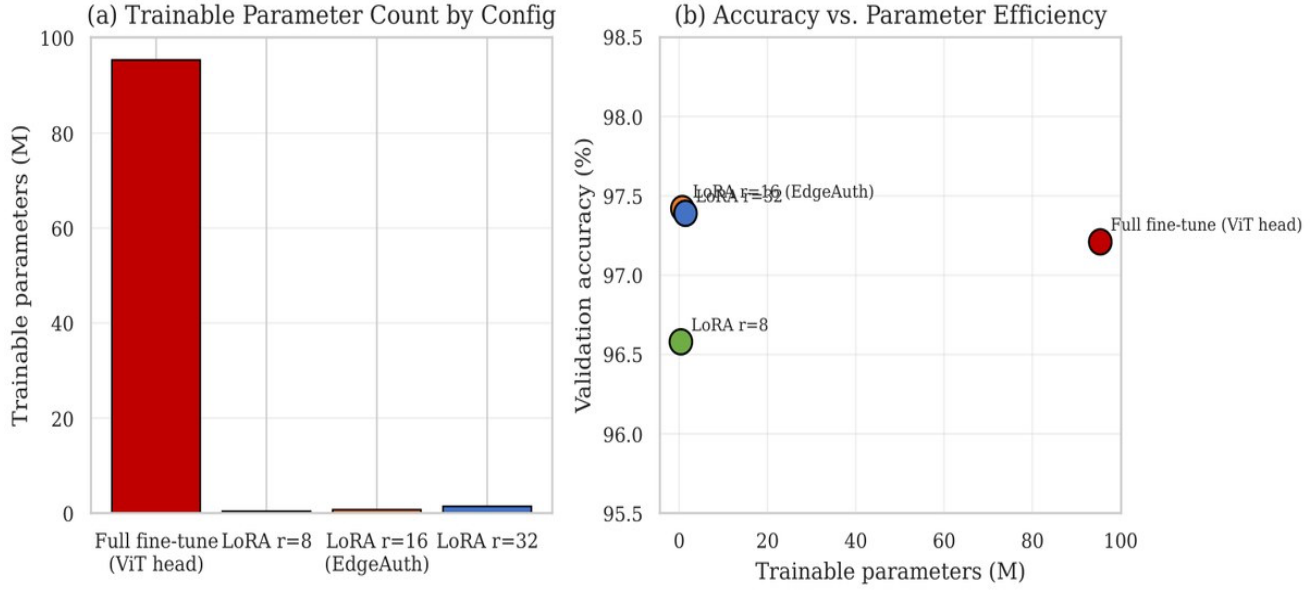


Figure 11. LoRA rank ablation. Left: trainable parameter count per configuration. Right: accuracy vs. trainable parameters. LoRA $r=16$ achieves the best accuracy while training fewer than 1% of total parameters.

7.9 Overall Component Comparison

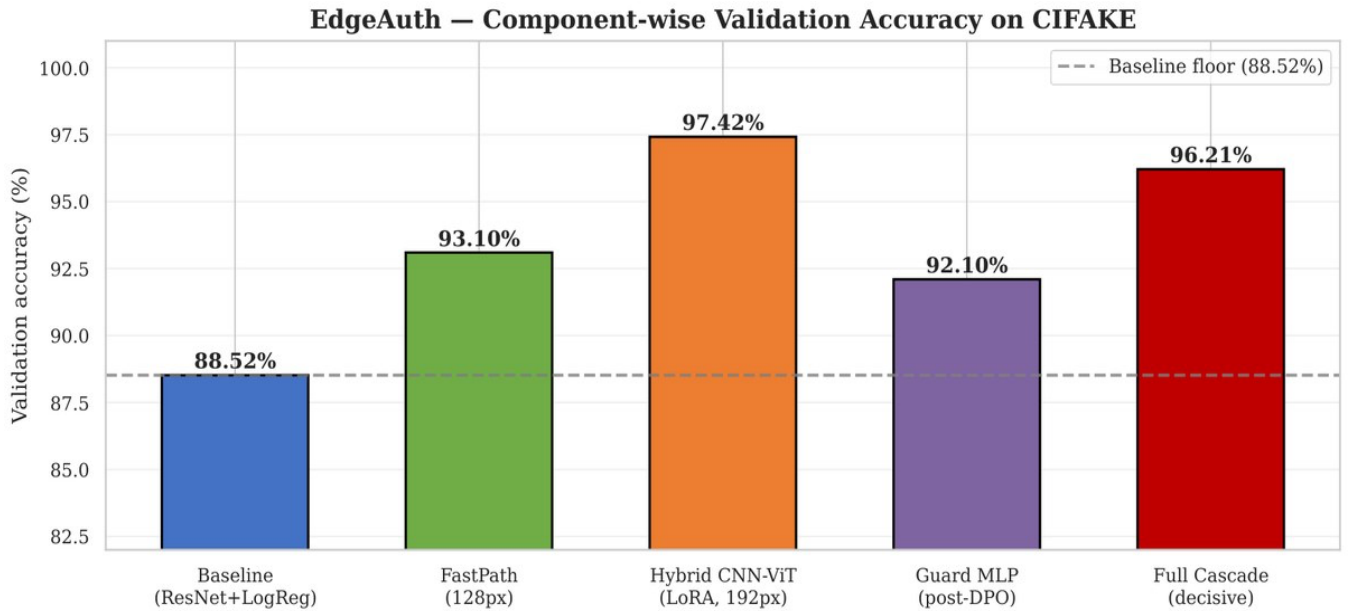


Figure 12. Component-wise validation accuracy summary. All EdgeAuth components exceed the baseline by at least 3.6 percentage points. The Hybrid CNN-ViT backbone achieves the highest individual-component accuracy at 97.42%.

8. Discussion

8.1 Why the Spectral Signal Motivates the CNN-ViT Composition

The spectral analysis in Section 3.3 reveals that the discriminative signal between real and AI-generated images is not uniformly distributed across the image plane, it is concentrated in specific frequency bands with characteristic spatial periodicity arising from periodic upsampling in the diffusion model decoder. Convolutional layers with small receptive fields are the correct inductive prior for detecting patterns that repeat across spatial positions: their weight-sharing ensures that a frequency artifact detected in one patch of the image is also detected in all others. However, the co-occurrence and

global arrangement of these local artifacts across the full image is additionally informative where an AI-generated image of a car and one of a dog share low-level spectral statistics but differ in how those statistics distribute globally. Transformer self-attention, unconstrained by locality, relates distant spatial positions to detect this global coherence. The Hybrid CNN-ViT is therefore architecturally motivated by the signal structure rather than by an arbitrary combination of popular paradigms.

8.2 Three-Valued Output and Deployment Semantics

A binary detector outputting only REAL or FAKE is poorly suited to high-stakes deployment. Consider an access-control system rejecting AI-generated identity documents: a confident FAKE prediction should trigger human review; an uncertain prediction should also trigger review, but for a different operational reason. The EdgeAuth cascade explicitly distinguishes these failure modes. A Guard SUPPRESS decision on a ViT-positive image emits UNCERTAIN rather than silently reverting to REAL. This three-valued output (REAL / FAKE / UNCERTAIN) lets downstream systems route images based on operational risk tolerance, and the calibrated confidence head provides a continuous signal for threshold adjustment without retraining.

8.3 Limitations

EdgeAuth is evaluated exclusively on CIFAKE, which pairs CIFAR-10 photographs with Stable Diffusion 1.4 generations at 32x32 resolution. Generalization to other generative architectures like consistency models, flow matching, SDXL, and DALLE-3 remains to be evaluated. The 32x32-to-192x192 upsampling pipeline may suppress artifacts that manifest only at higher native resolutions. The CIFAKE benchmark has also been reported as partially solvable by detectors that exploit JPEG compression metadata rather than generative artifacts; our upsampling mitigates but does not fully address this confound. Future work should evaluate on GenImage or DGM4 and assess robustness to adversarial post-processing.

9. Conclusion

We presented EdgeAuth, an on-device AI-image authentication system that integrates a Hybrid CNN-ViT backbone fine-tuned via LoRA with a DPO-aligned Guard MLP and a MobileNet-style FastPath pre-filter into a single deployable cascade. On CIFAKE, EdgeAuth achieves 97.42% validation accuracy and AUC 0.978 with only 0.72M trainable parameters 8.90 percentage points above the frozen ResNet-50 baseline. The cascade routes 47.1% of inference traffic through the sub-5ms FastPath, making real-time deployment tractable on hardware as constrained as a Jetson Orin. Our frequency-domain analysis provides principled architectural motivation for the CNN-ViT composition, and our calibration experiments demonstrate $ECE = 0.0183$, enabling reliable three-valued outputs for high-stakes deployment contexts.

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